Problems on the four operations with integers

Mixed calculations and word problems — where the sign rules meet the real world.

LESSON 33–34 | 2 × 40 MIN | MATEMATIKA 7, §14 | S8 1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Mixed expressions	15 min
The distributive law	25 min
Word problems	25 min
Averages	20 min
Homework	5 min

LEARNING OBJECTIVES

- Evaluate expressions containing all four operations and brackets.
- Translate a worded situation into an expression with signs.
- Use the distributive law with negative numbers.
- Check an answer for plausibility as well as for arithmetic.

TEXTBOOK

National	pp. 89–94
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1 review

01 Explanation

Mixed expressions

Work brackets first, then powers, then × and ÷, then + and −.

EXPRESSION	WORKING	VALUE
$-5 + 3 \times (-4)$	$-5 + (-12)$	-17
$(-5 + 3) \times (-4)$	$(-2)(-4)$	8
$-20 \div (-4) - 6$	$5 - 6$	-1
$(-3)^2 - 4 \times (-2)$	$9 + 8$	17

THE BRACKETS CHANGE EVERYTHING

The first two rows use the same three numbers and give -17 and 8. Copying an expression without its brackets is the most expensive mistake in this topic.

The distributive law

$$a(b + c) = ab + ac$$

It holds for negative numbers exactly as for positive ones, and it is how brackets are removed.

EXPRESSION	EXPANDED	VALUE
$-3(4 + 5)$	$-12 - 15$	-27
$-3(4 - 5)$	$-12 + 15$	3
$-(x - 7)$	$-x + 7$	—
$5 - (3 - 8)$	$5 - 3 + 8$	10

A MINUS IN FRONT OF A BRACKET CHANGES **EVERY** SIGN INSIDE

$-(x - 7) = -x + 7$, not $-x - 7$. Missing the second sign is the classic slip, and it will recur in every algebra lesson this year.

Word problems

SITUATION	EXPRESSION	ANSWER
$-7°$, rising $3°$ an hour for 4 hours	$-7 + 3 \times 4$	$5°$
a debt of $20\,000$, paid off at 6000 a month for 3 months	$-20\,000 + 3 \times 6000$	-2000
a diver at $-18\ m$ descending $4\ m$ a minute for 5 minutes	$-18 - 4 \times 5$	$-38\ m$
a lift on floor 12 going down 3 floors 4 times	$12 - 3 \times 4$	0

WRITE THE EXPRESSION BEFORE COMPUTING ANYTHING

The mathematics of these problems is easy; the translation is not. Getting the expression on paper first separates the two difficulties.

Averages

The **average** (arithmetic mean) of a list is the sum divided by how many there are.
With negatives it works exactly the same way.

$$mean = \frac{\text{sum of the values}}{\text{how many values}}$$

LIST	SUM	COUNT	MEAN
$-3, 0, 2, -5, 1$	-5	5	-1
$-8, -4, 0, 4$	-8	4	-2
$-6, 6$	0	2	0

THE MEAN ALWAYS LIES BETWEEN THE LEAST AND THE GREATEST

A mean of -1 for a list running from -5 to 2 is plausible; a mean of 3 would not be. That single check catches most arithmetic slips.

02 Worked examples

EXAMPLE 1 **Evaluate $-5 + 3 \times (-4)$ and $(-5 + 3) \times (-4)$.**

1. First: multiply before adding.
2. $-5 + (-12) = -17$
3. Second: bracket first.
4. $(-2)(-4) = 8$

   Same numbers, different answers.

ANSWER

-17 and 8

EXAMPLE 2 **The temperature is $-7°C$ and rises by 3 degrees each hour for 4 hours. Find it then.**

1. Expression: $-7 + 3 \times 4$.
2. $3 \times 4 = 12$
3. $-7 + 12$
4. $= 5°C$

   Now above zero.

ANSWER

$5°C$

EXAMPLE 3 **Find the mean of $-3, 0, 2, -5, 1$.**

1. Sum: $-3 + 0 + 2 - 5 + 1$.
2. $= -5$
3. Five values: $-5 \div 5$.
4. $= -1$ — between -5 and 2 ✓

ANSWER

-1

03 Interactive model

Give the same three numbers with and without brackets and let the class compute both; the difference makes the order of operations matter rather than merely exist.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Expression	Ifoda	Выражение
Brackets	Qavslar	Скобки
Distributive law	Taqsimot qonuni	Распределительный закон
To evaluate	Qiymatini topmoq	Вычислить
Word problem	Matnli masala	Текстовая задача
Average	O‘rta qiymat	Среднее
Net change	Sof o‘zgarish	Чистое изменение
Plausible	Ishonarli	Правдоподобный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$-5 + 3 \times (-4)$ equals:

- 8
- -17
- -32
- 32

$(-5 + 3) \times (-4)$ equals:

- 8
- -17
- -8
- 2

$-(x - 7)$ equals:

- $-x - 7$
- $-x + 7$
- $x - 7$
- $x + 7$

$5 - (3 - 8)$ equals:

- 0
- 10
- -6
- 16

The mean of $-8, -4, 0, 4$:

- -2
- -8
- 0
- 2

A mean always lies:

- above the greatest
- below the least
- between them
- at zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up `7 PROBLEMS`

1. $-5 + 3 \times (-4)$

   ANSWER -17

2. $(-5 + 3) \times (-4)$

   ANSWER 8

3. $-20 \div (-4) - 6$

   ANSWER -1

4. $(-3)^2 - 4 \times (-2)$

   ANSWER 17

5. $-3(4 + 5)$

   ANSWER -27

6. $-3(4 - 5)$

   ANSWER 3

7. $5 - (3 - 8)$

   ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Mean of $-3, 0, 2, -5, 1$

   ANSWER -1

2. Mean of $-8, -4, 0, 4$

   ANSWER -2

3. $-7 + 3 \times 4$

   ANSWER 5

4. $-18 - 4 \times 5$

   ANSWER -38

5. $-20\,000 + 3 \times 6000$

   ANSWER -2000

6. Simplify $-(a - b + c)$

   ANSWER $-a + b - c$

7. $(-2)(-3) - (-4)(5)$

   ANSWER 26

Hard · stretch and reason

7 PROBLEMS

1. $(-4)^2 - (-4)^2 \div (-2)^3$

   ANSWER 18

2. $-2[3 - (-5 + 1)]$

   ANSWER -14

3. Mean of $-12, -5, 0, 3, 9$

   ANSWER -1

4. A diver at -12 m rises 2 m a minute for 8 minutes

   ANSWER $+4$ m

5. Five days at $-4, -1, 2, -7, 5$: the mean temperature

   ANSWER $-1°$

6. Find x: $3x - (-6) = 0$

   ANSWER -2

7. Evaluate $a^2 - 2ab$ at $a = -3, b = 4$

   ANSWER 33

07 Homework

Homework — 5 tasks

Write the expression before the arithmetic, and check the answer for plausibility.

1. Evaluate $-6 + 4 \times (-3)$ and $(-6 + 4) \times (-3)$.
2. Evaluate $(-2)^3 + 5 \times (-4) \div 2$.
3. Simplify $-(2x - 5 + y)$.
4. The temperature is $-9°C$ and rises 4 degrees each hour for 3 hours. Find it then.
5. Find the mean of $-10, -2, 0, 4, 8$.